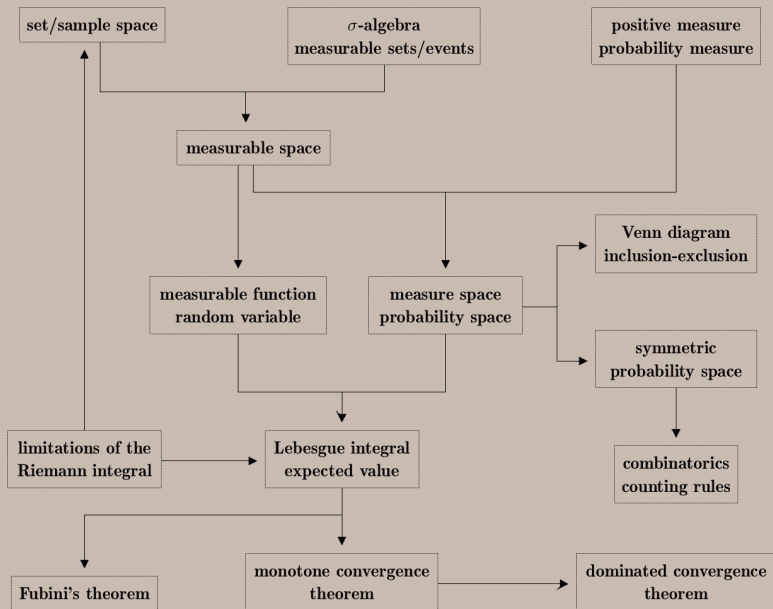
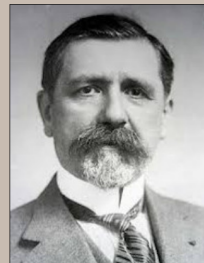
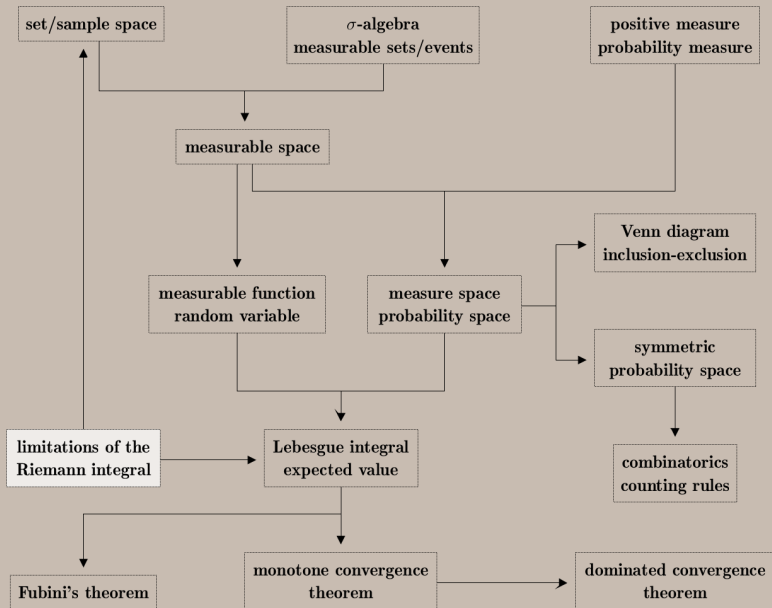


Emile Borel



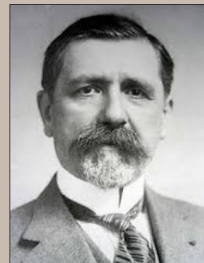
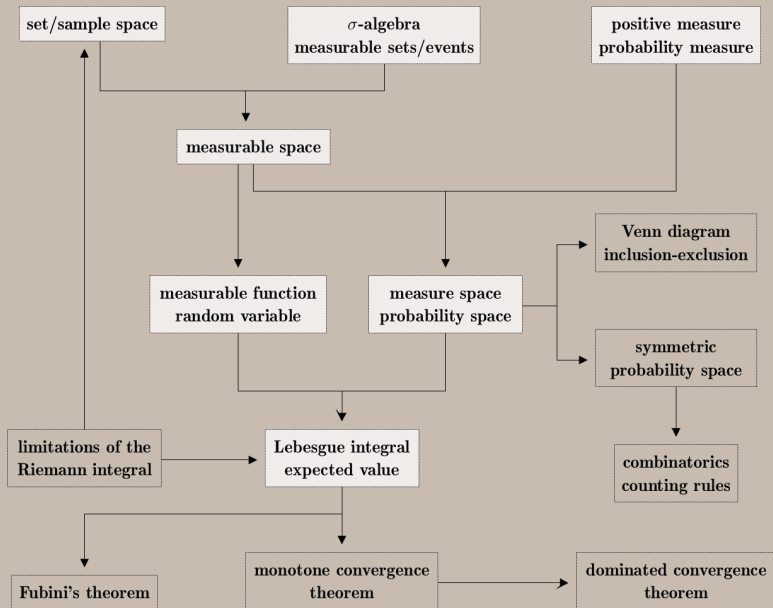
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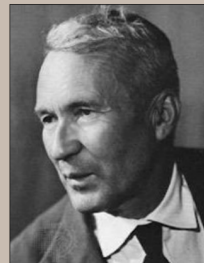
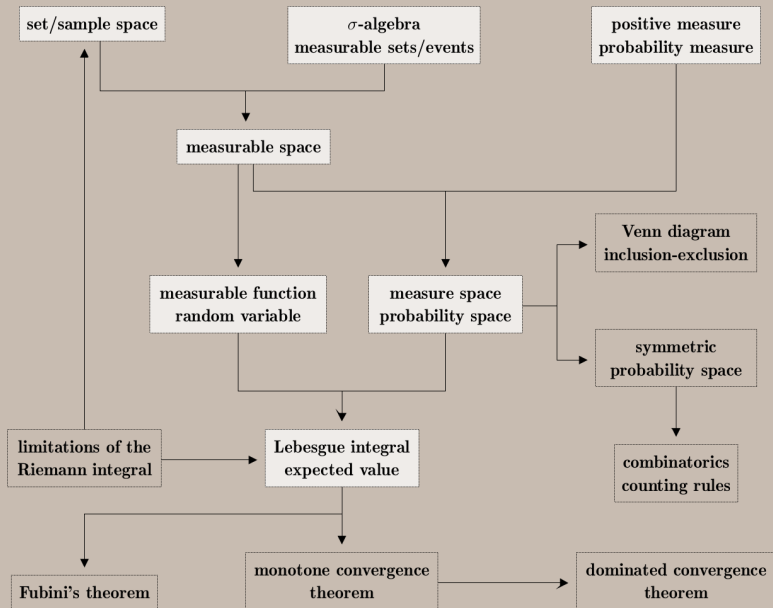
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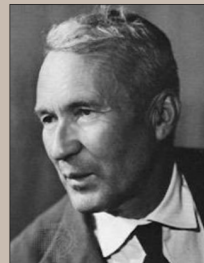
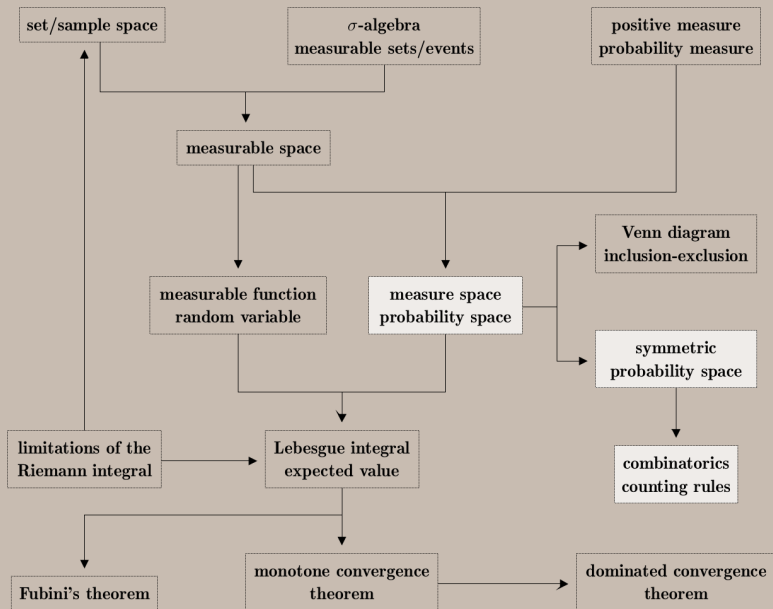
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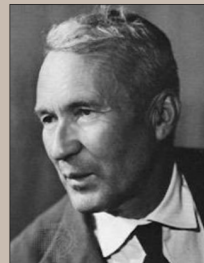
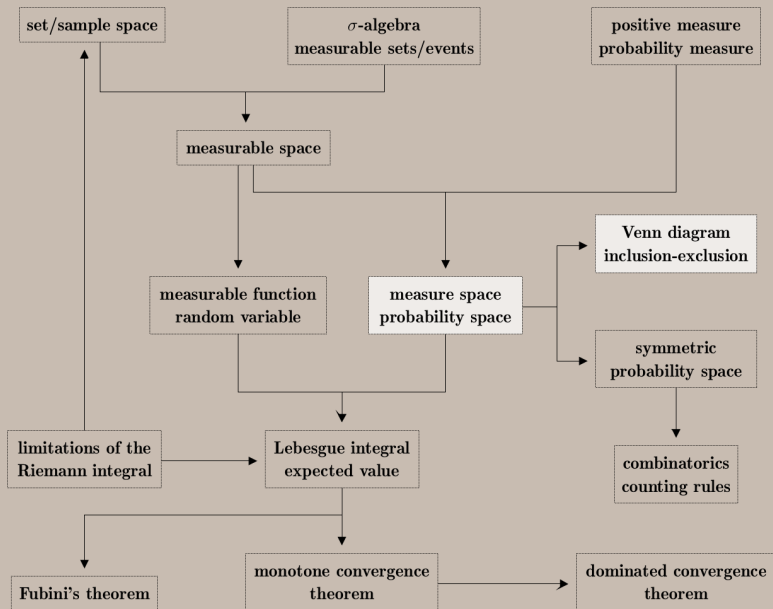
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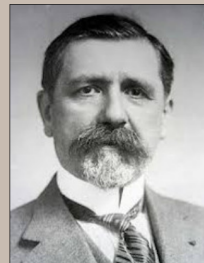
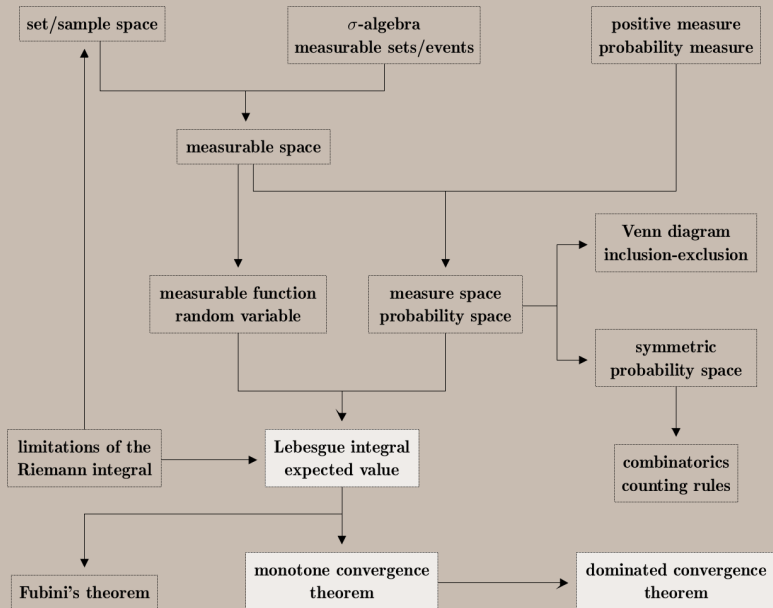
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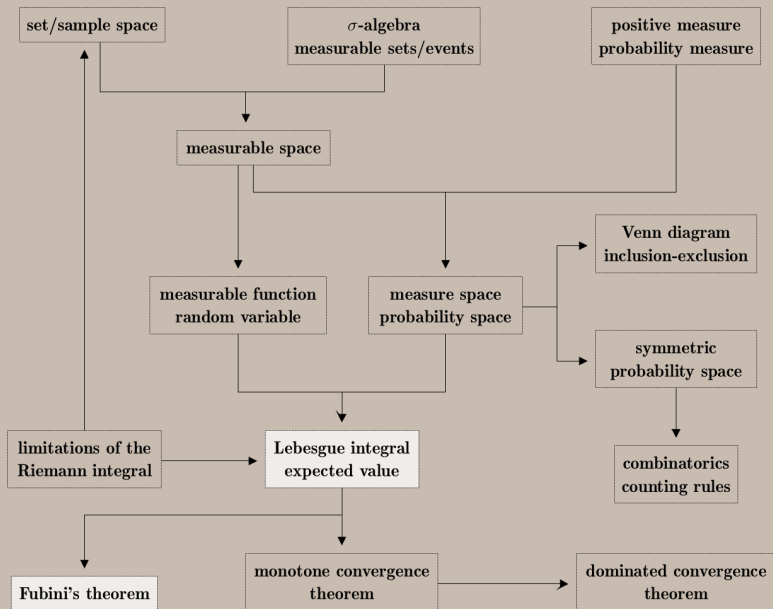
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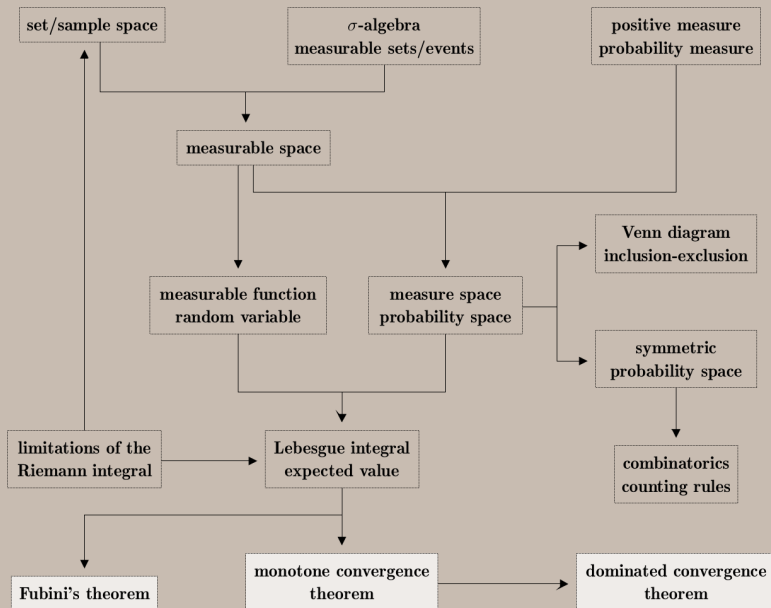
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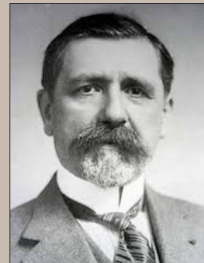
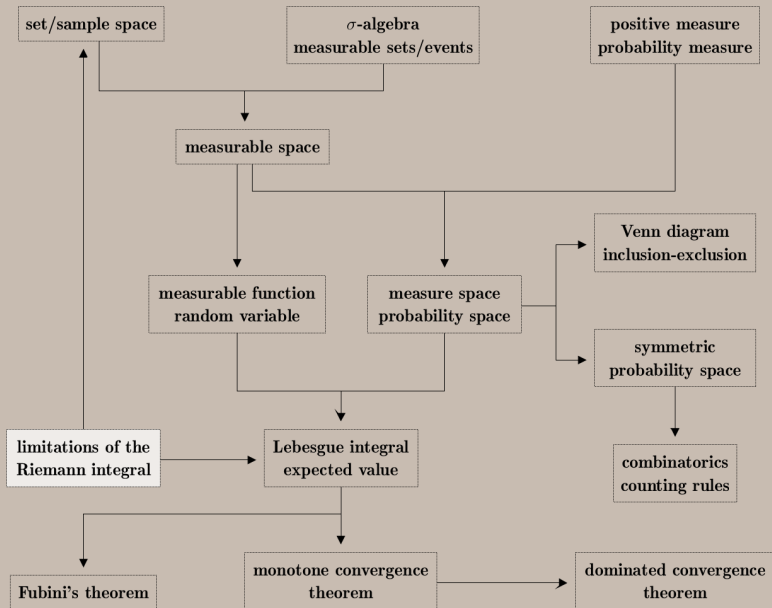
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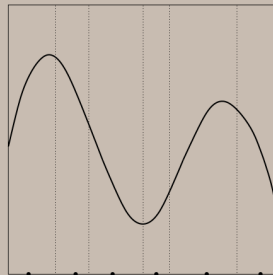
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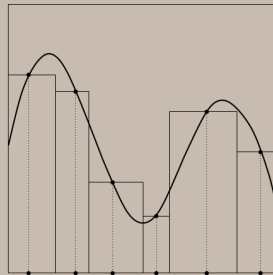
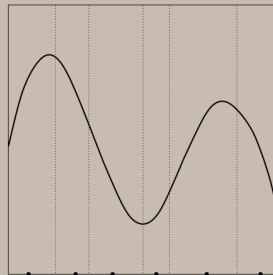


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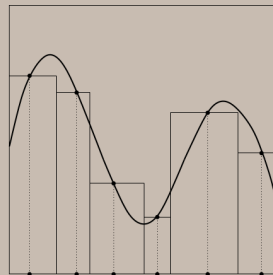
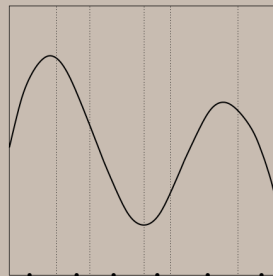
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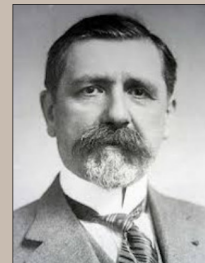
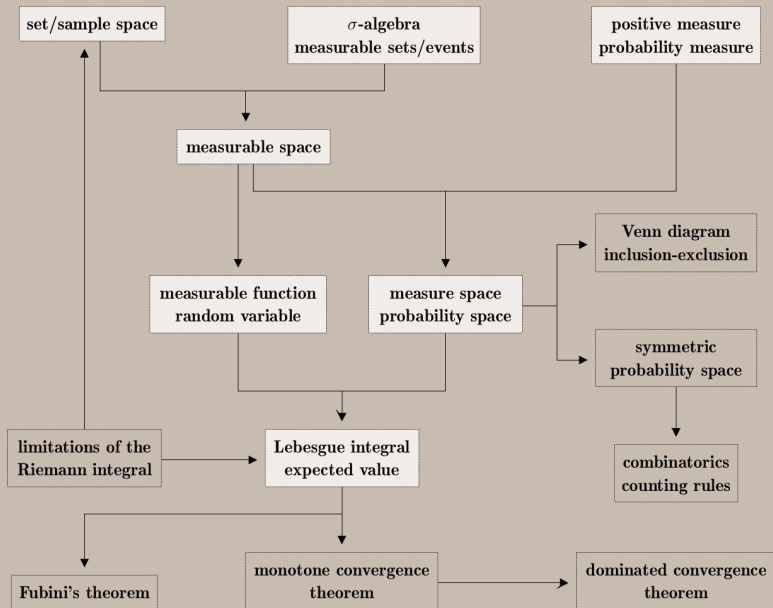
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which does not exist: $\mathbf{1}_{\mathbb{Q}}$ is not Riemann integrable.





Emile Borel



Henri Lebesgue

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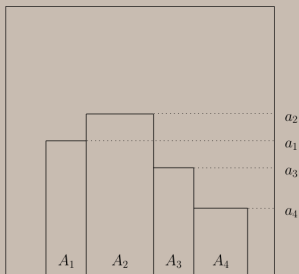
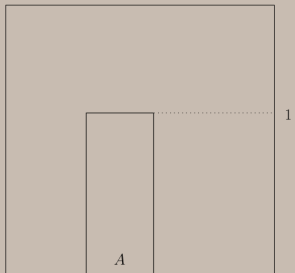
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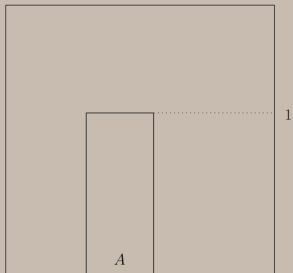
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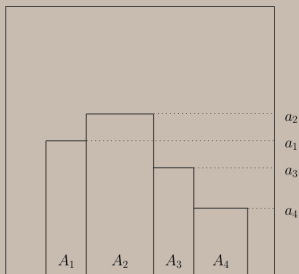
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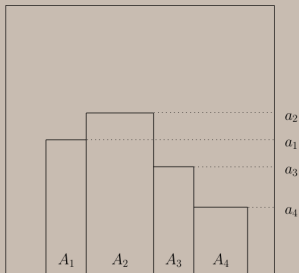
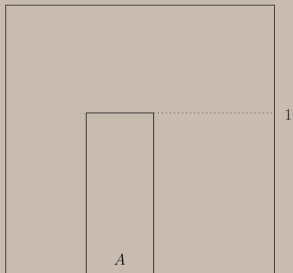
Example. **Lebesgue measure λ** = unique measure on $\mathcal{B}$ with $\lambda((a, b)) = b - a$ for all $a < b$.





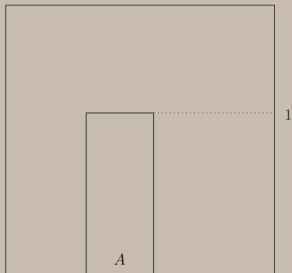
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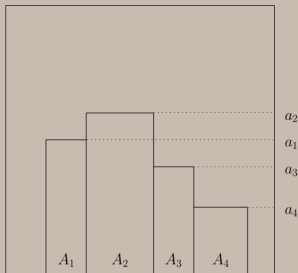
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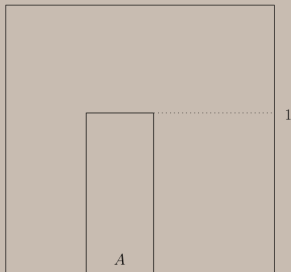


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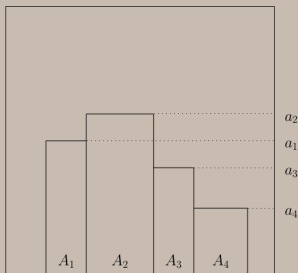
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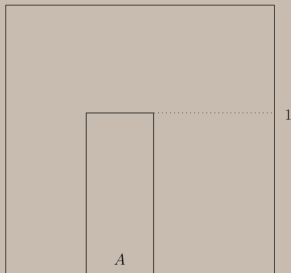
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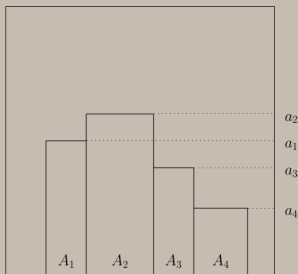
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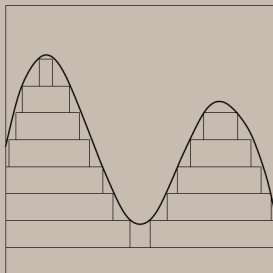
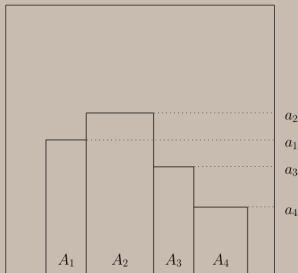
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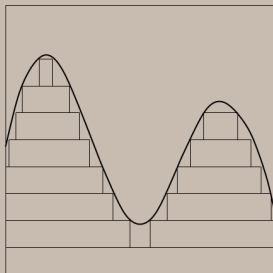
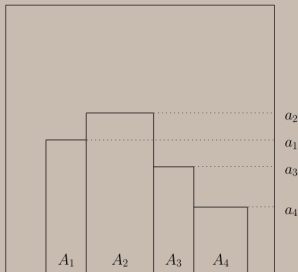
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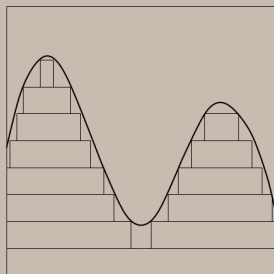
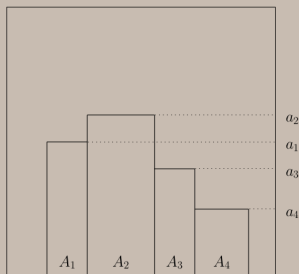
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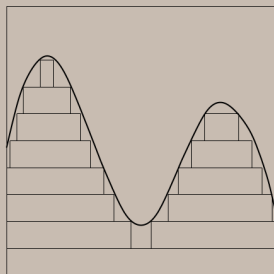
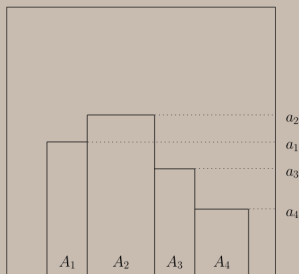
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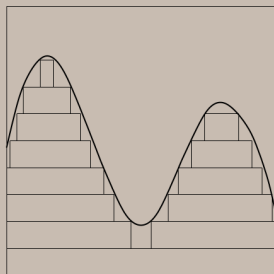
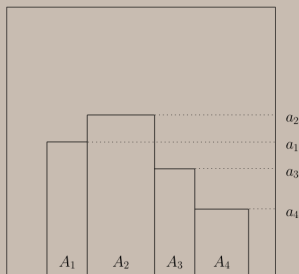
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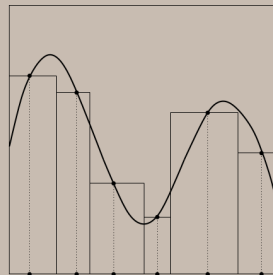
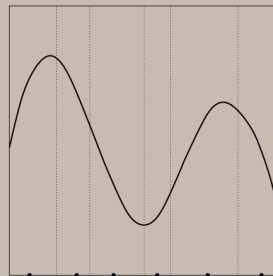
What is the probability that a number chosen uniformly at random in the unit interval is a rational number?

$X = \text{Uniform}(0, 1)$. Density function: $\phi = \mathbf{1}_{(0,1)}$.

The probability to be found is

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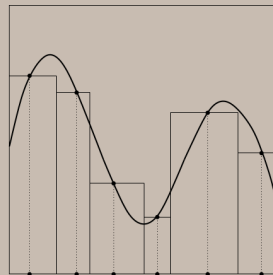
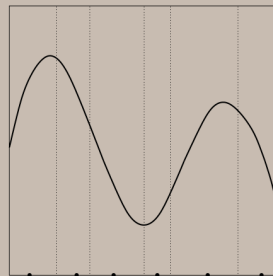
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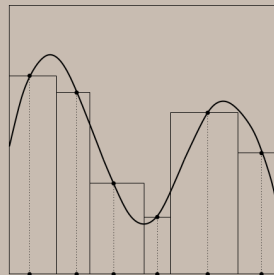
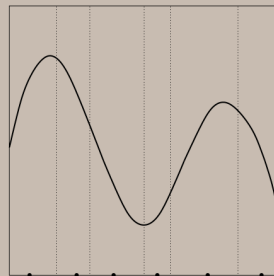
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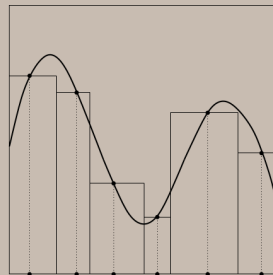
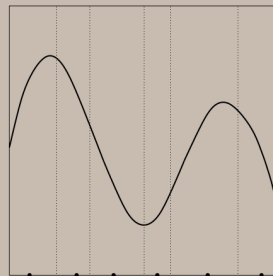
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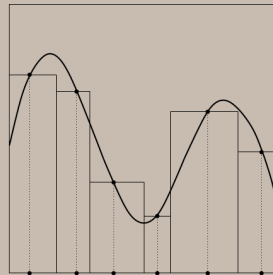
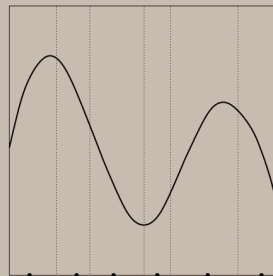
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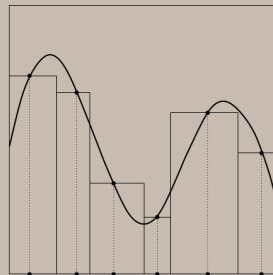
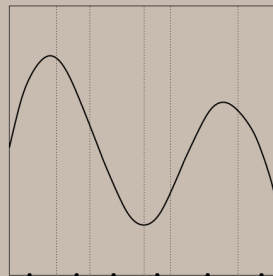
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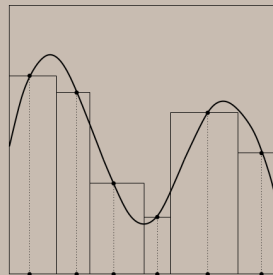
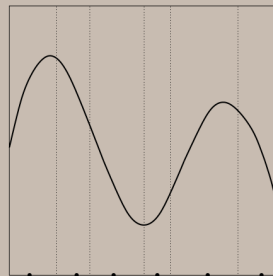
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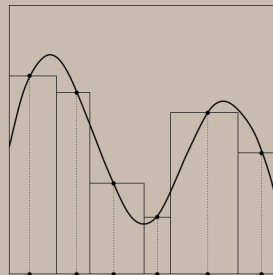
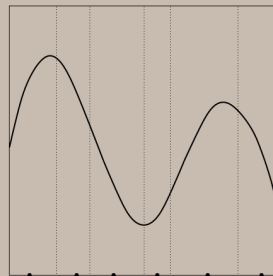
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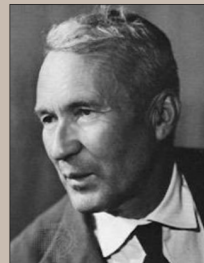
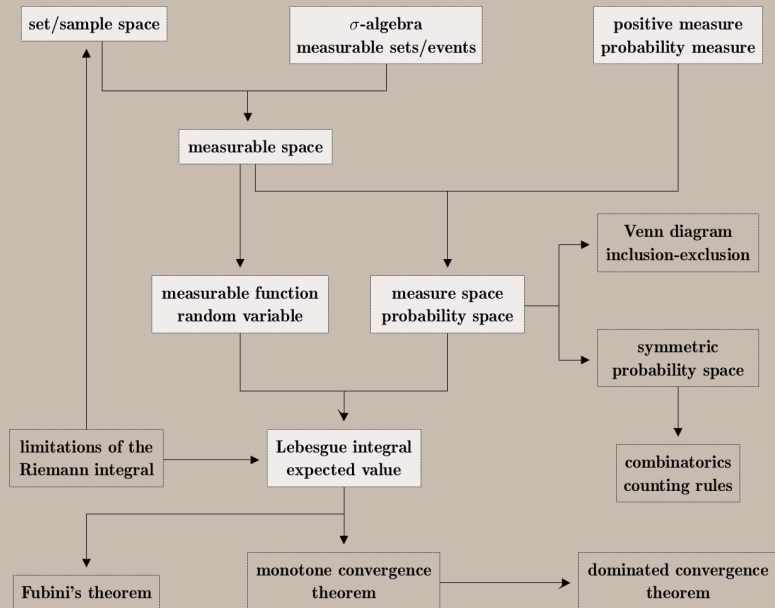
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Andrey Kolmogorov



Henri Lebesgue

Definition. A collection $\mathcal{F}$ of subsets of Ω is a **σ -algebra** whenever

$$(1) \Omega \in \mathcal{F} \quad (2) A \in \mathcal{F} \implies A^c \in \mathcal{F} \quad (3) (A_n) \subset \mathcal{F} \implies \cup_n A_n \in \mathcal{F}.$$

Note: $(1) + (2) \implies \emptyset = \Omega^c \in \mathcal{F}$, $(2) + (3) \implies \cap_n A_n = (\cup_n A_n^c)^c \in \mathcal{F}$, etc.

The pair $(\Omega, \mathcal{F})$ is a **measurable space**, and sets in $\mathcal{F}$ are **measurable sets**.

Definition. $\sigma(\mathcal{H}) =$ **σ -algebra generated by $\mathcal{H}$** = smallest σ -algebra containing $\mathcal{H}$.

Example. **Borel σ -algebra** on the real line: $\mathcal{B} = \sigma\{(a, b) : a, b \in \mathbb{R} \text{ and } a < b\}$.

Note: $(2) + (3) \implies \{x\} = ((-\infty, x) \cup (x, \infty))^c \in \mathcal{B} + (3) \implies \mathbb{Q} = \cup_{x \in \mathbb{Q}} \{x\} \in \mathcal{B}$.

Definition. $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B})$ is a **measurable function** if

$$X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\} = \{X \in B\} \in \mathcal{F} \quad \text{for all } B \in \mathcal{B}.$$

Definition. A function $\mu : \mathcal{F} \rightarrow [0, \infty]$ is a **positive measure** if it is σ -additive:

$$\mu(\cup_n A_n) = \sum_n \mu(A_n) \quad \text{for all } (A_n) \subset \mathcal{F} \text{ mutually exclusive.}$$

The triplet $(\Omega, \mathcal{F}, \mu)$ is a **measure space**.

Example. **Lebesgue measure** λ = unique measure on $\mathcal{B}$ with $\lambda((a, b)) = b - a$ for all $a < b$.

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Definition. $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B})$ is a measurable function $\rightarrow$ **random variable** if

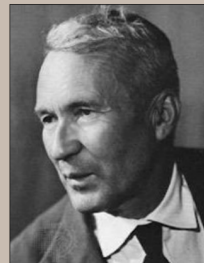
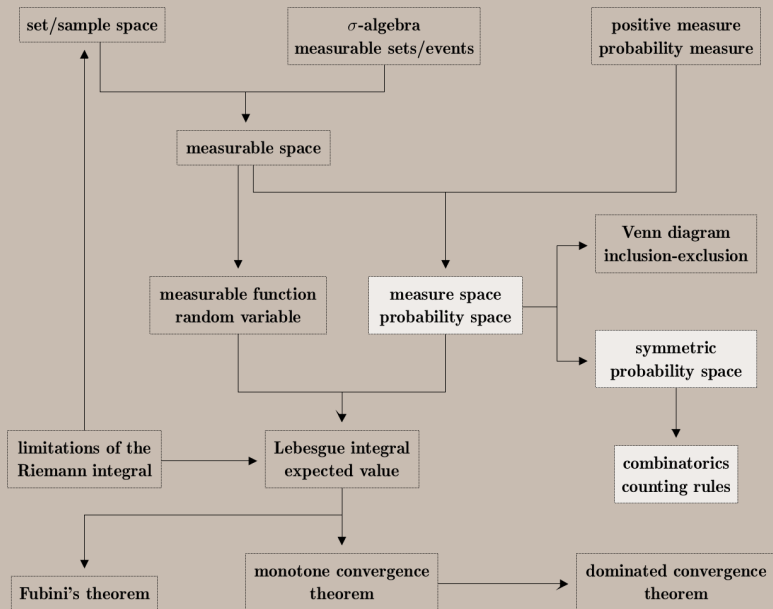
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Definition. A function $P : \mathcal{F} \rightarrow [0, 1]$ is a positive measure $\rightarrow$ **probability measure** if

$$P(\Omega) = 1 \quad \text{and} \quad P(\cup_n A_n) = \sum_n P(A_n) \quad \text{for all } (A_n) \subset \mathcal{F} \text{ mutually exclusive.}$$

The triplet $(\Omega, \mathcal{F}, P)$ is a measure space $\rightarrow$ **probability space**.

Definition. The **expected value** of a random variable X is the integral of X with respect to P .

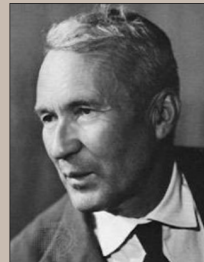


Andrey Kolmogorov



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Symmetric probability space: Sample space Ω finite, and all outcomes equally likely (fair coin, fair die, well-mixed deck of cards, etc.).



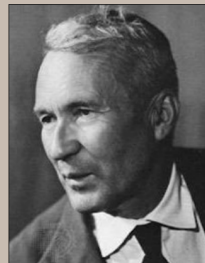
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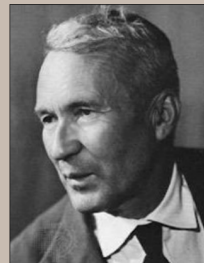
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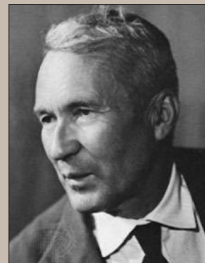
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For every event $A \in \mathcal{F}$,

$$P(A) = P(\cup_{\omega \in A} \{\omega\}) \stackrel{\text{ADD}}{=} \sum_{\omega \in A} P(\{\omega\}) = p|A| = |A|/|\Omega|.$$



Andrey Kolmogorov



Henri Lebesgue

Symmetric probability space: Sample space Ω finite, and all outcomes equally likely (fair coin, fair die, well-mixed deck of cards, etc.).

Let $P(\{\omega\}) = p$ for all $\omega \in \Omega$.

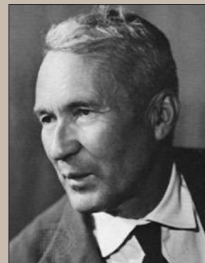
$$p|\Omega| = \sum_{\omega} P(\{\omega\}) \stackrel{\text{ADD}}{=} P(\cup_{\omega} \{\omega\}) = P(\Omega) = 1$$

$$\implies P(\{\omega\}) = 1/|\Omega| \text{ for all } \omega \in \Omega.$$

For every event $A \in \mathcal{F}$,

$$P(A) = P(\cup_{\omega \in A} \{\omega\}) \stackrel{\text{ADD}}{=} \sum_{\omega \in A} P(\{\omega\}) = p|A| = |A|/|\Omega|.$$

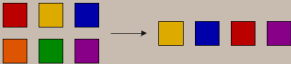
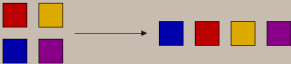
Goal: count the number of outcomes in $|A|$ and $|\Omega|$.

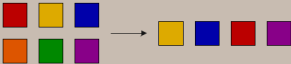
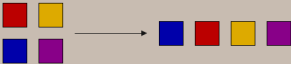


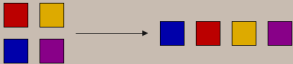
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	Arrangement choose a subset and order its elements	Permutation order the elements of a given set	Combination choose a subset of a given set
without repetition	Number of orderings of k blocks chosen from a set of n distinct blocks 	Number of orderings of n distinct blocks 	Number of subsets of k blocks chosen from a set of n distinct blocks 
with repetition	Number of orderings of k blocks each of which can be of n different colors 	Number of orderings of $n = n_1 + \dots + n_k$ blocks with n_i blocks of color i 	Number of subsets of k blocks each of which can be of n different colors 

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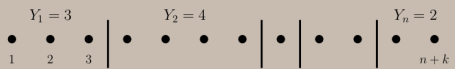


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$Y_1 = 3$

$\bullet_1 \quad \bullet_2 \quad \bullet_3$

$|$

$Y_2 = 4$

$\bullet \quad \bullet \quad \bullet$

$|$

$\bullet$

$|$

$\bullet$

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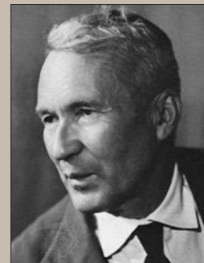
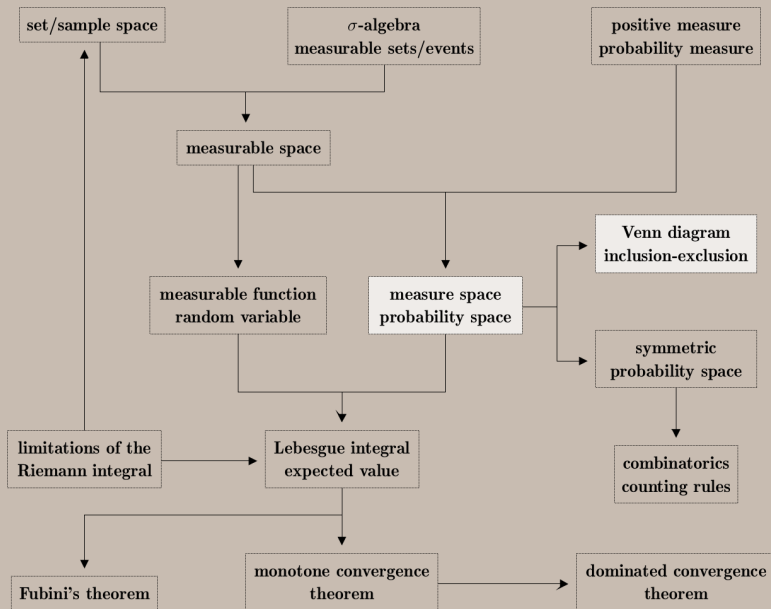
$Y_n = 2$

$\bullet \quad \bullet$

$n+k$

$\left(\begin{matrix} n+k-1 \\ n-1 \end{matrix} \right) = \frac{(n+k-1)!}{k! (n-1)!}$

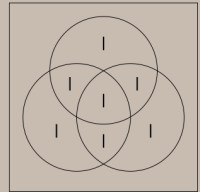
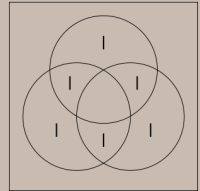
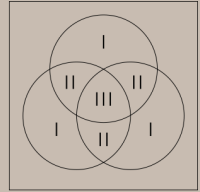
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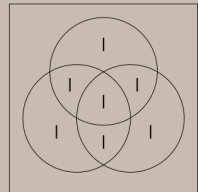
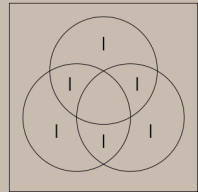
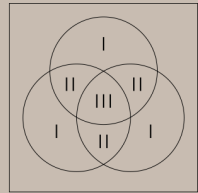
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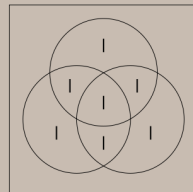
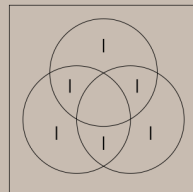
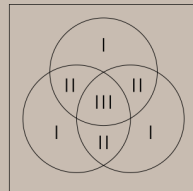


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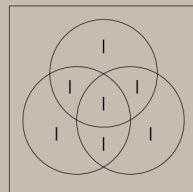
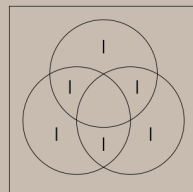
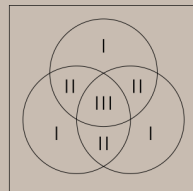


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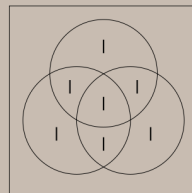
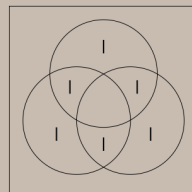
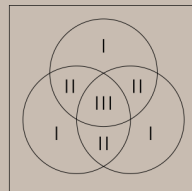
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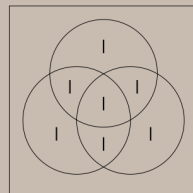
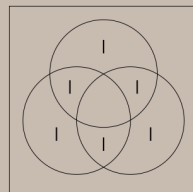
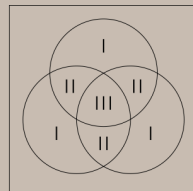
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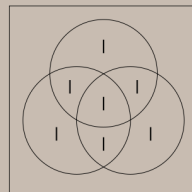
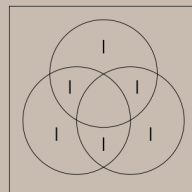
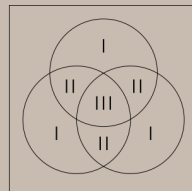
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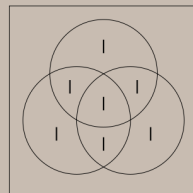
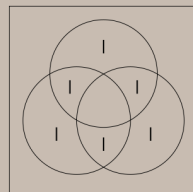
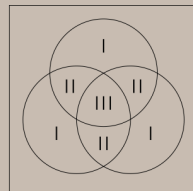
$$P(\cup_{i=1}^n A_i) = \sum_{k=1}^n (-1)^{k+1} \sum_{i_1 < \dots < i_k} P(A_{i_1} \cap \dots \cap A_{i_k}).$$

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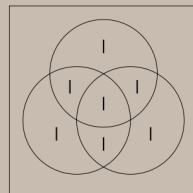
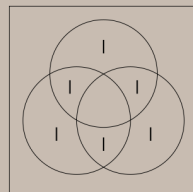
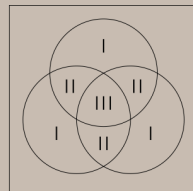
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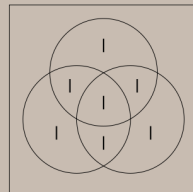
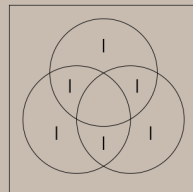
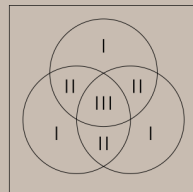
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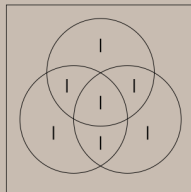
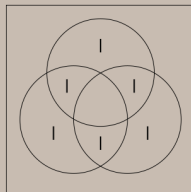
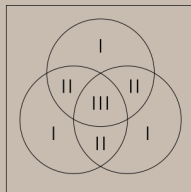
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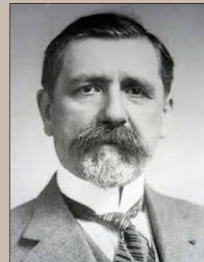
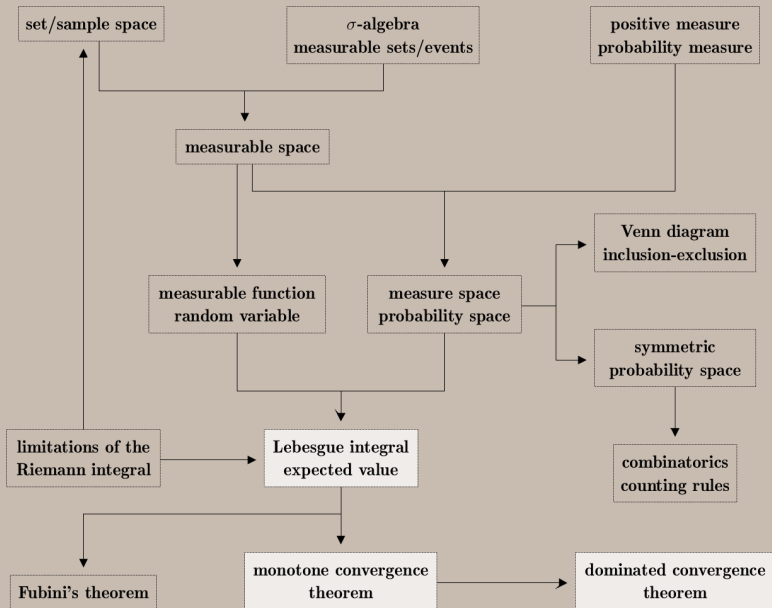
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Emile Borel



Henri Lebesgue

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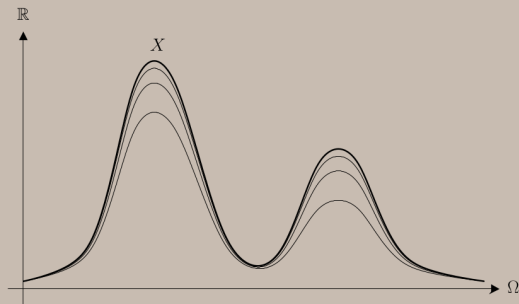
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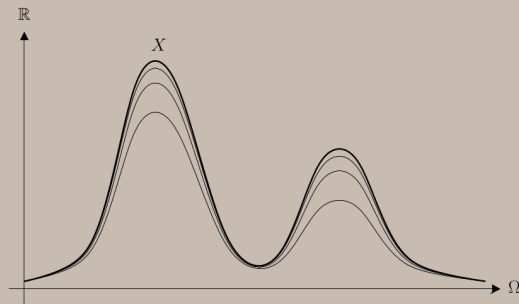
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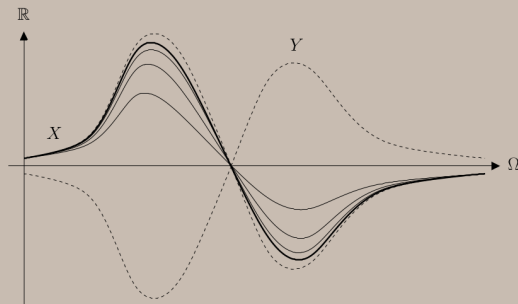
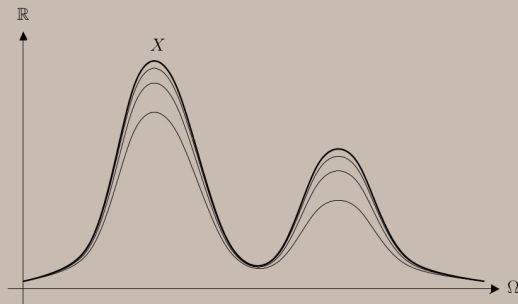
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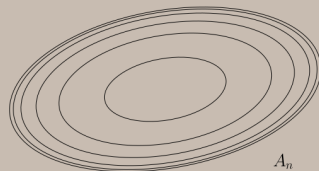
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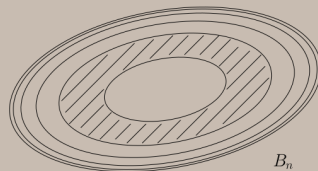
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A diagram of a disk B_n in the complex plane. It consists of several concentric ellipses. The region between the innermost ellipse and the next ellipse out is shaded with diagonal lines. The label B_n is placed at the bottom right of the diagram.

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Proof. Apply Fatou's lemma to $2Y - |X_n - X|$.

Lemma. $(A_n) \subset \mathcal{F}$ nondecreasing $\implies \nu(\cup_n A_n) = \lim_{n \rightarrow \infty} \nu(A_n)$.

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Fatou's lemma. Let $X_n \geq 0$. Then, $\int \liminf_{n \rightarrow \infty} X_n d\mu \leq \liminf_{n \rightarrow \infty} \int X_n d\mu$.

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$$\begin{aligned} \int 2Y d\mu &= \int \liminf_{n \rightarrow \infty} (2Y - |X_n - X|) d\mu \stackrel{\text{FL}}{\leq} \liminf_{n \rightarrow \infty} \int (2Y - |X_n - X|) d\mu \\ &= \int 2Y d\mu - \limsup_{n \rightarrow \infty} \int |X_n - X| d\mu \leq \int 2Y d\mu. \end{aligned}$$

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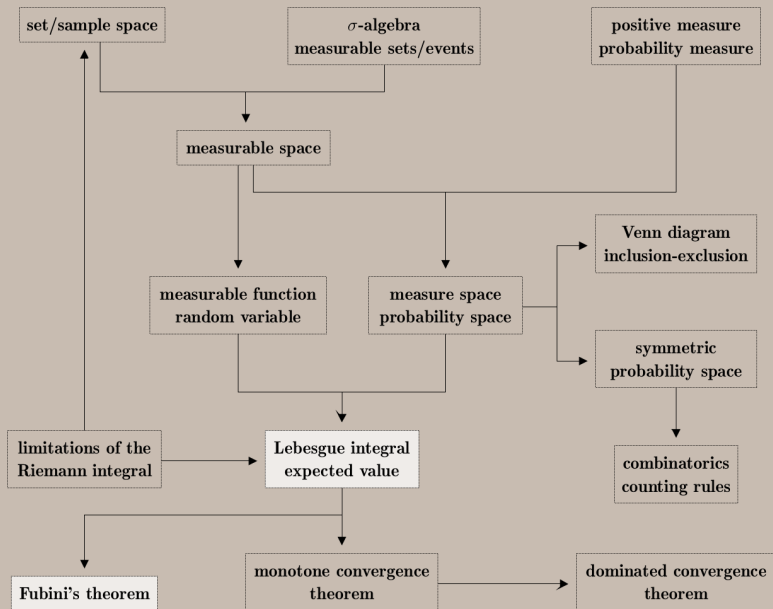
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This implies that

$$\begin{aligned} \limsup_{n \rightarrow \infty} \left| \int X_n d\mu - \int X d\mu \right| &\leq \limsup_{n \rightarrow \infty} \|X_n - X\|_1 \\ &= \limsup_{n \rightarrow \infty} \int |X_n - X| d\mu = 0. \end{aligned}$$



Guido Fubini



Henri Lebesgue

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Definition. The measure space $(\Omega, \mathcal{F}, \mu)$ is **σ -finite** if Ω can be written as a countable union of sets with finite measure.



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Definition. The measure space $(\Omega, \mathcal{F}, \mu)$ is **σ -finite** if Ω can be written as a countable union of sets with finite measure.

Fubini's theorem. Let $X : (\Omega, \mathcal{F}, \mu) \rightarrow (\mathbb{R}, \mathcal{B})$ be a positive measurable function or an integrable function on a σ -finite measure space. Then,

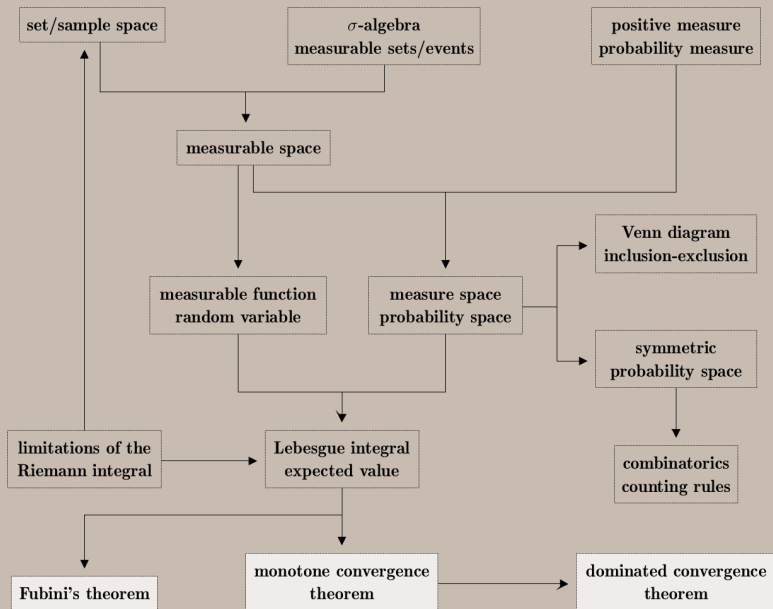
$$\int_{\Omega_2} \int_{\Omega_1} X d\mu_1 d\mu_2 = \int_{\Omega_1} \int_{\Omega_2} X d\mu_2 d\mu_1 = \int_{\Omega} X d\mu.$$



Guido Fubini



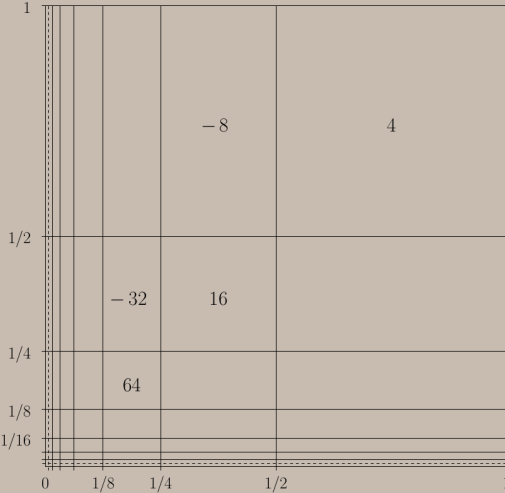
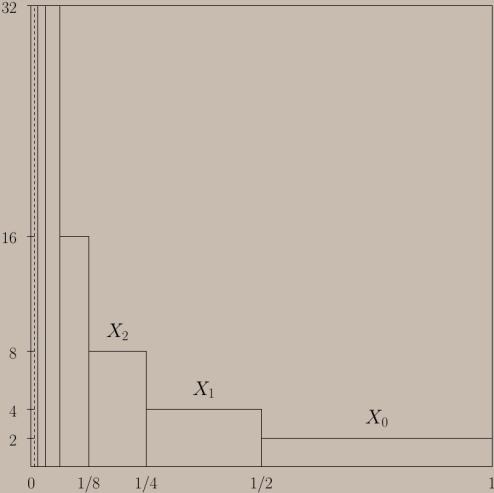
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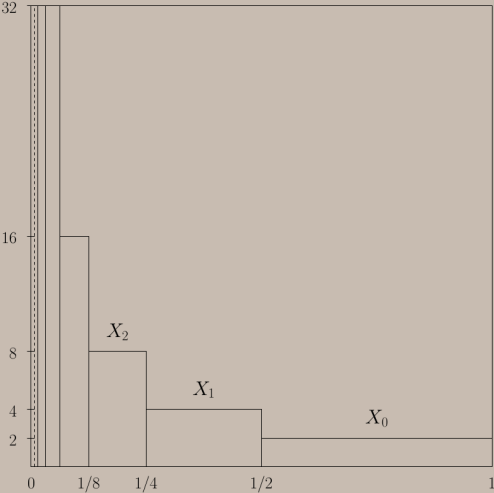


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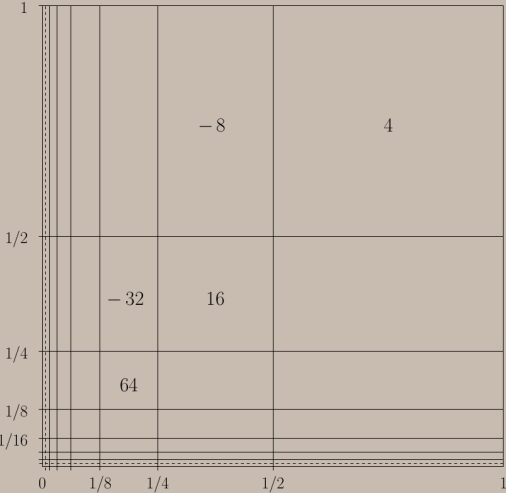


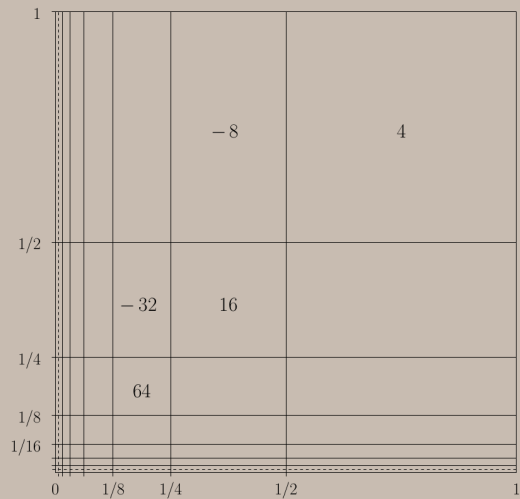
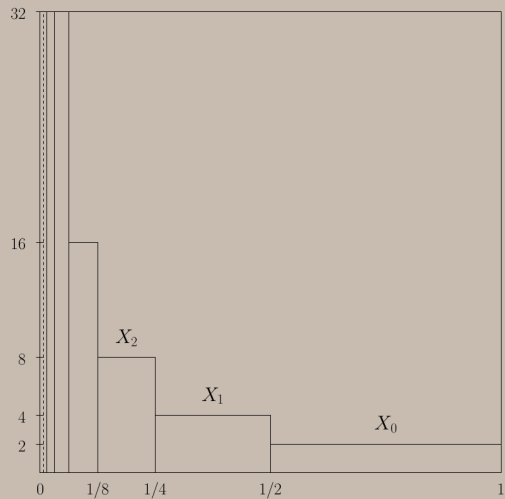
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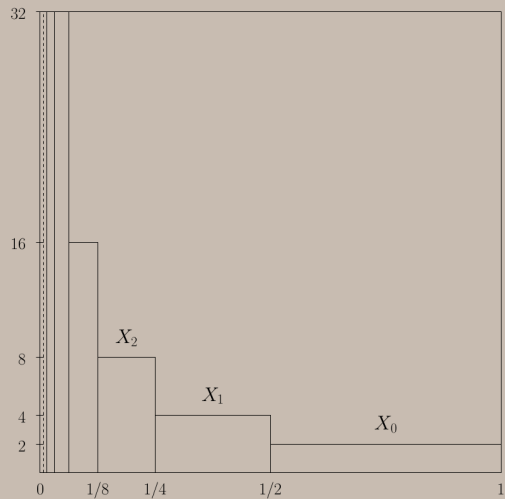


$$\lim_{n \rightarrow \infty} \int X_n d\lambda = 1$$

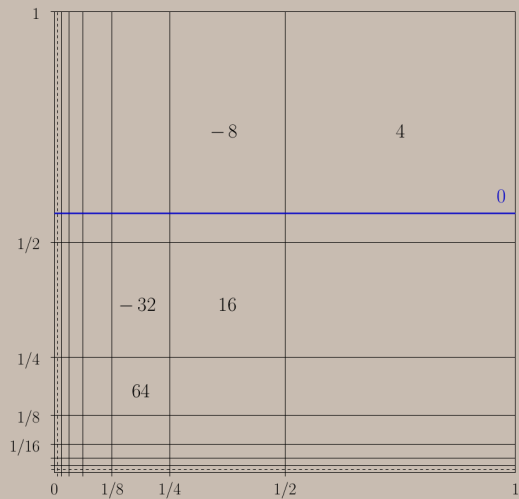


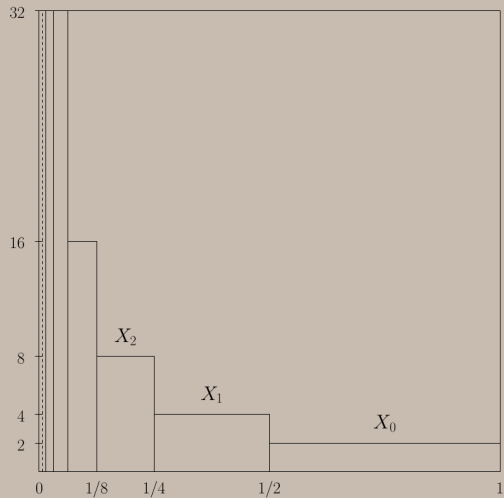


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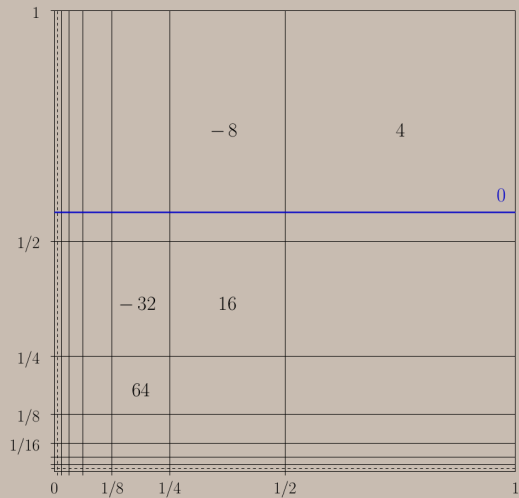


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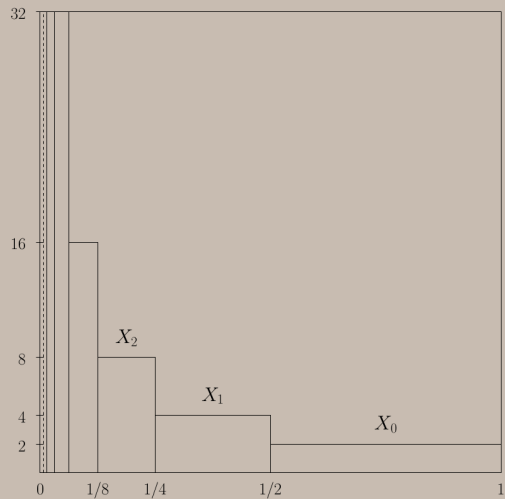




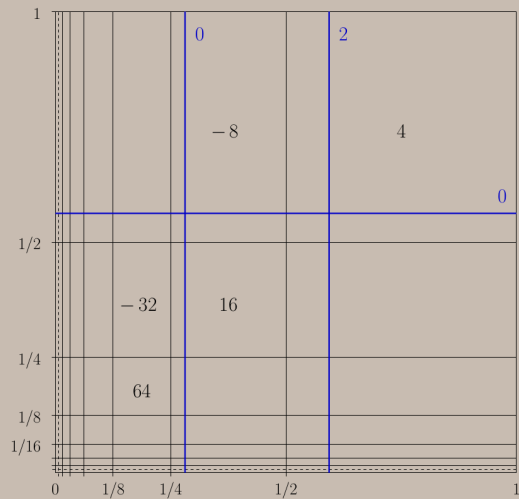
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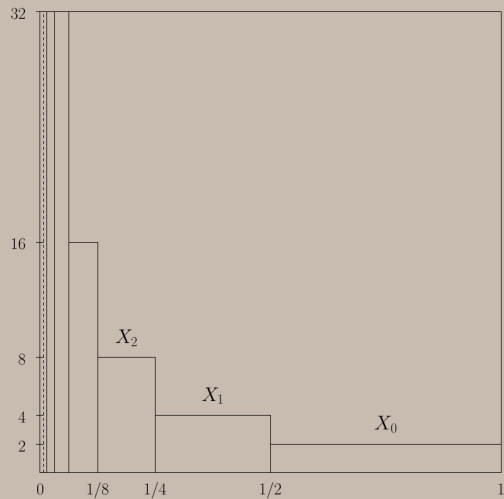
$$\iint X(\omega) d\omega_1 d\omega_2 = 0$$



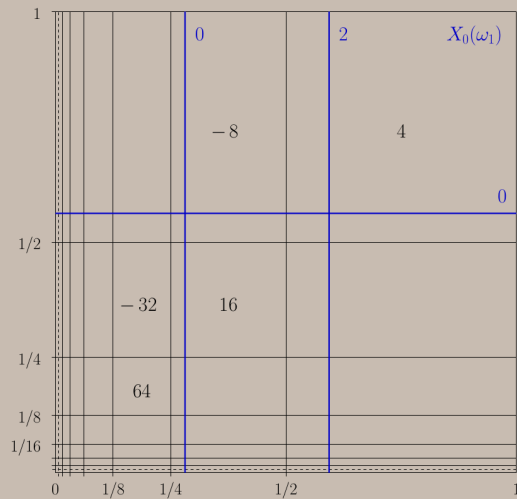
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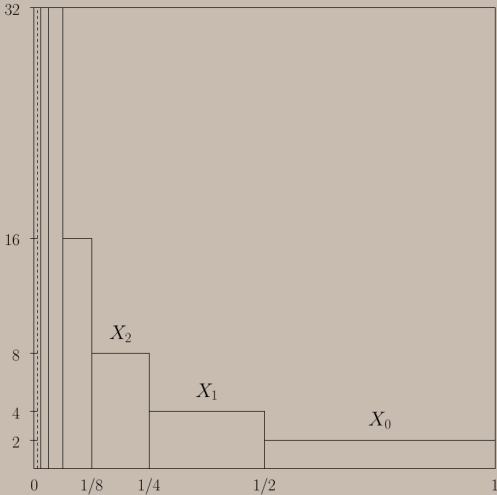
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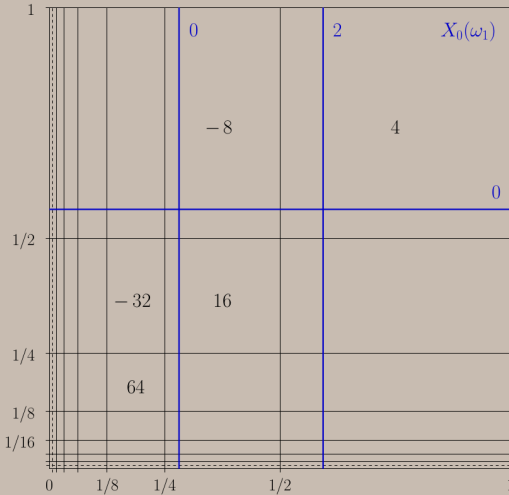
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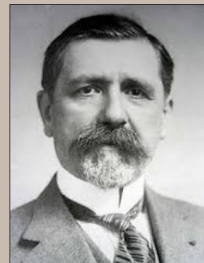
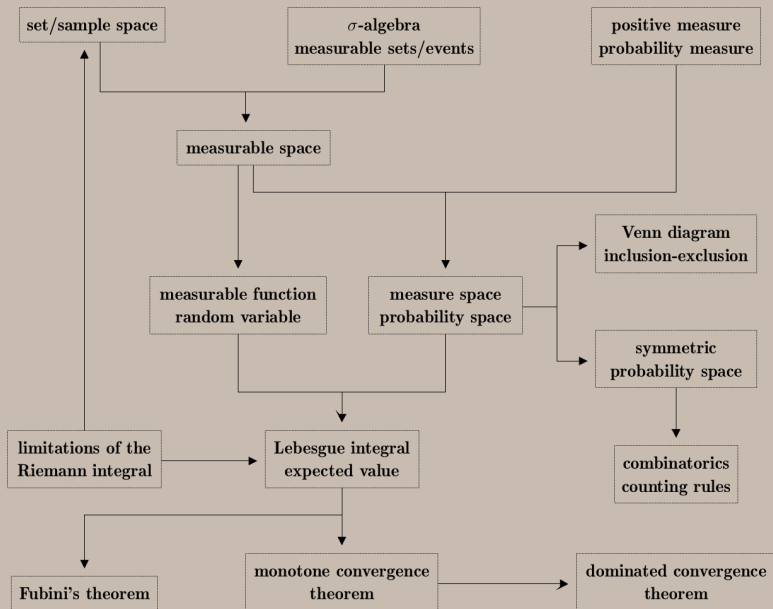
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Emile Borel



Henri Lebesgue